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Faculty of Science & Technology (FST)
Department of Computer Science

FINAL-TERM PROJECT REPORT
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Project Title	Student Performance Analysis
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Course / Section	Programming In Python [Sec: D]
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1. Executive Summary

This project investigates the academic performance of secondary school students using machine learning classification techniques. The dataset, sourced from Kaggle, contains 1,044 student records across 34 attributes covering demographic, social, and academic variables. The primary objective was to predict whether a student passes or fails based on these attributes, excluding their actual grade scores. Three models were trained and evaluated: a K-Nearest Neighbor (KNN), a Support Vector Machine (SVM), and a Random Forest classifier. To address a significant class imbalance in the target variable, SVM and RF were trained with balanced class weights. Random Forest achieved the highest overall accuracy at 79.9% (baseline model), followed by KNN at 77% and SVM at 72.3%. The results indicate that behavioral and demographic features carry meaningful predictive signal, though moderate accuracy levels reflect the inherent difficulty of predicting academic outcomes without grade history.

2. Introduction and Project Objectives

Student academic performance is influenced by a complex mix of social, familial, and behavioral factors. Understanding which of these factors are most predictive of failure can help educational institutions intervene early and allocate support resources more effectively. This project applies binary classification to predict student pass or fail outcomes using survey-based features that exclude direct grade information.

Item	Your Content
Dataset name	Student Performance Analysis
Dataset source	https://www.kaggle.com/datasets/safianaam/student-performance
Main project goal	Predict whether a student passes or fails their final exam using demographic, social, and behavioral features
Research Question 1	Which classification model performs better on this imbalanced dataset?
Research Question 2	How does class imbalance affect model performance and what steps can mitigate it?

3. Dataset Description

The dataset contains records of secondary school students from two Portuguese schools. Each row represents one student and includes information about their school, family background, study habits, social behavior, and academic history. The target variable is derived from the final grade G3.

3.1 Basic Dataset Information

Property	Value
Number of rows	1,044
Number of columns	34 (original), reduced to 27 after dropping irrelevant columns

Data types present	Numerical (integer) and categorical (object)
Target or key variable(s)	pass (1) vs fail (0); derived from G3
Potential issues observed	Significant class imbalance: 78% Pass, 22% Fail

3.2 Initial Observations

Upon first inspection, the dataset contained no missing values and no duplicate rows, which reduced the need for imputation. However, a clear class imbalance was observed in the target variable: 814 students passed (78%) and only 230 failed (22%). Several columns such as nursery attendance, home address, family size, parental cohabitation status, reason for school selection, and alcohol consumption were identified as weakly relevant to academic performance and were removed before modelling. The grade columns G1, G2, and G3 were also handled carefully. G3 was converted into the binary target variable, while G1 and G2 were dropped to prevent data leakage.

4. Data Cleaning and Preparation

Issue Found	Action Taken	Reason / Justification
7 irrelevant columns present (nursery, address, famsize, Pstatus, reason, Dalc, Walc)	Dropped using df.drop()	These columns have weak theoretical relevance to academic performance and add noise to the model
Grade columns G1 and G2 present alongside target G3	Dropped before modelling	G1 and G2 are intermediate grades that directly leak information about G3, artificially inflating model accuracy
Significant class imbalance in target (78% Pass, 22% Fail)	Applied class_weight= 'balanced'	Prevents models from defaulting to majority class prediction; ensures the minority class (Fail) receives appropriate attention during training

5. Feature Engineering

New Feature	How It Was Created	Why It Is Useful
pass(1) vs fail(0)	G3 >= 10 mapped to 1 (Pass), below 10 mapped to 0 (Fail)	Converts a continuous grade into a binary classification target, enabling supervised classification
Binary encoded columns (schoolsup, famsup, paid, activities, higher, internet,	yes/no values mapped to 1 and 0; F/M mapped to 0 and 1	Converts categorical yes/no responses into numerical format required by machine learning

romantic, sex)		models
One-hot encoded columns (school, Mjob, Fjob, guardian, course)	pd.get_dummies() applied to multi-class categorical columns, expanding them into separate binary columns	Prevents the model from assuming any ordinal relationship between unordered categorical values

After encoding, the feature matrix expanded from 27 columns to 36 features, as multi-class categorical columns were split into individual binary columns. The final dataset used for modelling contained 1,044 rows and 36 features.

6. Analysis Methodology

The analysis followed a structured machine learning pipeline implemented in Python using Pandas, NumPy, and scikit-learn, with Matplotlib used for all visualizations.

- Pandas was used for loading the dataset, inspecting data types and distributions, dropping irrelevant columns, encoding categorical variables, and constructing the feature matrix and target vector.
- NumPy was used for converting the prepared data frame into arrays suitable for scikit-learn models, and for computing scaling statistics displayed after standardization.
- Scikit-learn was used for the train/test split with stratification, StandardScaler for feature scaling, and for training and evaluating all three classification models. Evaluation metrics included accuracy, precision, recall, and F1 score computed using weighted averaging to account for class imbalance.
- Matplotlib was used for all visualizations including confusion matrices for each model and the grouped bar chart comparing all metrics across models.

7. Analysis and Visualizations

7.1 Research Question 1

Which model performs best?

Random Forest (baseline model) achieved the highest overall accuracy at 79.9% compared to KNN at 77% and SVM at 72.3%. In terms of precision, SVM scored 75.3%, slightly higher than Random Forest at 77.2% and KNN at 70.9%. For F1 score, Random Forest led at 76.3%, followed by SVM at 73.5% and KNN at 71.4%. The confusion matrices for each model reveal the trade-off between correctly identifying Pass and Fail students.

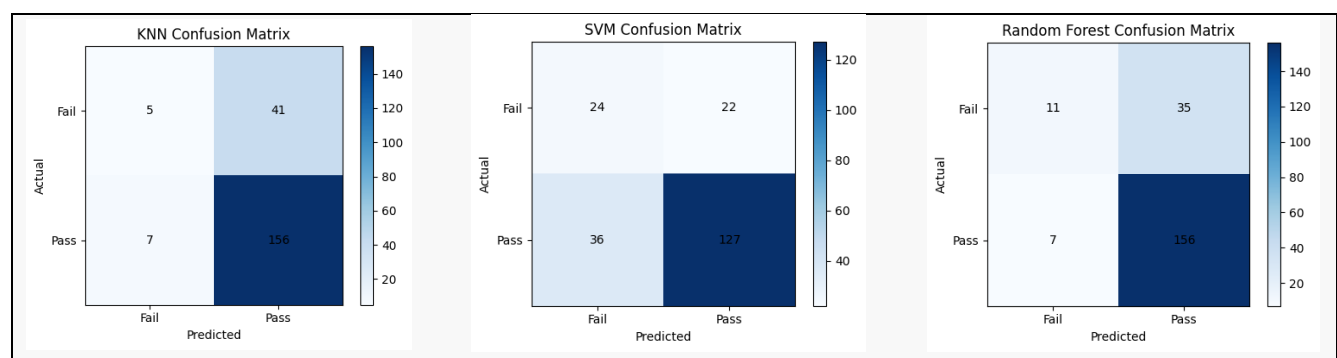


Figure 1. Confusion matrix showing predicted vs actual pass/fail outcomes for all 3 models.

7.2 Research Question 2

How does class imbalance affect performance and what mitigates it?

Without addressing class imbalance, a model trained on this data would be biased toward predicting Pass for almost every student, since 78% of the dataset is Pass. By applying `class_weight='balanced'` to SVM and Random Forest, the training process penalized misclassification of the minority Fail class more heavily. The comparison bar chart below shows all four metrics side by side, making it clear that F1 score and recall provide a more complete picture of model performance than accuracy alone in imbalanced settings.

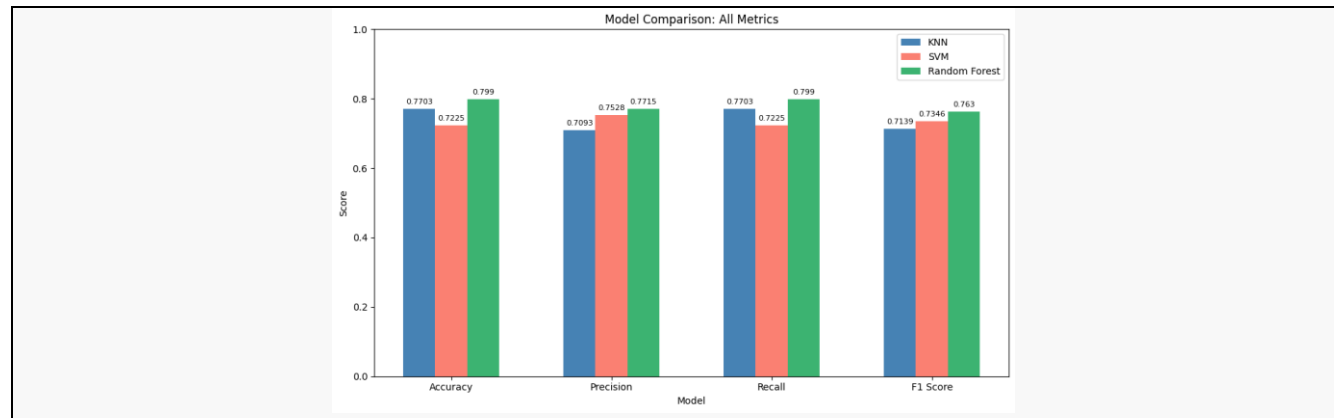


Figure 2. Grouped bar chart comparing Accuracy, Precision, Recall, and F1 Score across KNN, SVM, and Random Forest.

8. NumPy-Based Custom Computation

Component	Description
Computation used	Feature matrix conversion to NumPy arrays and post-scaling statistical verification
Purpose	To prepare the data in the format required by scikit-learn models and verify that standardization was applied correctly
Formula or logic	<code>df[feature_columns].to_numpy(dtype=np.float32)</code> converts the dataframe to a numerical array. After scaling, <code>X_train_scaled.mean(axis=0)</code> and <code>X_train_scaled.std(axis=0)</code> verify that all features have mean ≈ 0 and std ≈ 1
Result / takeaway	Post-scaling verification confirmed correct standardization across all 36 features, with means approximately 0 and standard deviations of 1, ensuring no feature dominates due to scale differences

9. Key Findings

1. The dataset contains a significant class imbalance - 78% of students passed and 22% failed. Without corrective measures, any model trained on this data would be biased toward predicting Pass, making accuracy a misleading metric on its own.
2. Random Forest achieved the highest performance across all metrics: accuracy 79.9%, precision 77.2%, recall 79.9%, and F1 score 76.3% making it the strongest overall model in this project.
3. KNN achieved 77.0% accuracy and 71.4% F1 score without any class imbalance handling, performing competitively but showing lower precision at 70.9%, indicating more false positives compared to the other models.
4. SVM with balanced class weights achieved the second highest precision at 75.3% and F1 score of 73.5%, demonstrating better handling of the minority Fail class compared to KNN despite having the lowest overall accuracy at 72.3%.
5. All three models outperformed what a naive majority-class predictor would achieve, confirming that demographic, behavioral, and school-related features contain meaningful predictive information about student academic outcomes even in the absence of prior grade data.

10. Limitations

The dataset is limited to students from two Portuguese schools, which restricts the generalizability of findings to other educational systems or countries. The class imbalance, despite being partially addressed through balanced class weights in SVM and Random Forest, still affects KNN's ability to detect failing students. Features are largely self-reported survey responses, which may contain response bias. The removal of G1 and G2 while necessary to prevent data leakage, significantly reduces predictive power, since prior academic performance is one of the strongest known predictors of final grades. Finally, the models identify statistical associations between features and outcomes but cannot establish causation; factors such as internet access or family support likely reflect underlying socioeconomic conditions not fully captured in the dataset.

11. Conclusion

This project applied binary classification to predict student academic outcomes using the Student Performance Analysis dataset. After removing irrelevant and leakage-prone columns, encoding categorical features, and addressing class imbalance through balanced class weights, three models were trained and evaluated- KNN, SVM, and Random Forest. Random Forest achieved the highest overall accuracy at 79.9% with an F1 score of 76.3%, making it the strongest performer across all metrics. KNN performed competitively at 77.0% accuracy without class weighting, while SVM achieved the lowest accuracy at 72.3% but demonstrated stronger precision at 75.3% due to balanced class weight handling. The results confirm that non-grade behavioral and demographic features can predict student outcomes with moderate reliability, while also highlighting that accuracy alone is insufficient as an evaluation metric when class distributions are unequal. F1 score and per-class recall provide a complete and more honest picture of model performance in such settings.

12. Dataset Source

<https://www.kaggle.com/datasets/safianaam/student-performance>